

CURRICULUM VITAE

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Research Assistant Professor

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RESEARCH INTERESTS

1. Subseasonal-to-Seasonal Prediction and Predictability

- Predictability, forecast skill, and systematic biases in coupled prediction systems
- Evaluation and improvement of predictions across weather-to-climate timescales

2. Climate Dynamics, Teleconnections, and Extremes

- Tropical climate variability and tropical–extratropical interactions
- Large-scale atmospheric circulation, teleconnection processes, and climate extremes

3. Coupled Earth System Processes and Data Assimilation

- Atmosphere–ocean and land–atmosphere interactions
- Coupled data assimilation and initialization of Earth system prediction models

EDUCATION

2015.03 – 2021.02	Ph.D. in Environmental Science and Engineering , UNIST, Korea Dissertation: <i>Development of a Coupled Data Assimilation System in the Fully Coupled Model and Its Implications for Seamless Prediction</i> Advisor: Prof. Myong-In Lee
2009.03 – 2015.02	B.S. in Earth Science and Engineering , UNIST, Korea

EMPLOYMENTS

2025.09 - Present	Research Assistant Professor , UNIST
2023.06 – 2025.09	Postdoctoral Research Fellow , George Mason University
2021.03 – 2023.06	Postdoctoral Research Associate , UNIST

RESEARCH EXPERIENCE

2023.01 – 2023.05	Visiting Scientist , Met Office Hadley Centre, UK
2017.02	Visiting Researcher , NASA GSFC, Climate and Radiation Lab, USA
2013.06 – 2013.08	Internship , NASA GSFC, GMAO, USA

PROFESSIONAL SERVICES

2024.10 – 2026.10	Editor , Korean Atmospheric Scientists in America (KASA)
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PUBLICATIONS

- [19] Lee, B., Seo, E., Kim, D., **Choi, N.**, & Moon, W. (2026). Understanding the physical mechanism of record-breaking 2025 Korea-Japan heatwave. *Environmental Research Letters*, 21, 194020.
- [18] Lee, S., Lee, M.-I., Kim, K.-M., & **Choi, N.** (2026). Extratropical Rossby waves amplify central Siberian fire risk through subseasonal teleconnections. *Agricultural and Forest Meteorology*, 390, 111475.
- [17] **Choi, N.**, Quan, X-W., and Stan, C. (2026). Drivers of the Asymmetric ITCZ Bias in Climate Models, *Scientific Reports*.
- [16] Lee, J., Lee, M. I., Kim, J., Bak, G., **Choi, N.**, Lee, J., ... & Shin, B. (2026). Atlantic SST as a key source of subseasonal predictability in the April 2020 Siberian heatwave. *npj Natural Hazards*.
- [15] **Choi, N.**, Stan, C., Lee, B., & Seo, E. (2026). Potential sources of predictability for the surface air temperature over the CONUS during boreal winter. *Journal of Climate*, 39(16), 5005-5016.
- [14] **Choi, N.**, Stanczak, J., & Stan, C. (2026). The Role of Atmosphere–Ocean Coupling on the Prediction of Madden–Julian Oscillation. *Journal of Climate*, 39(13), 3669-3684.
- [13] Kim, K., Lee, M. I., Lee, S., **Choi, N.**, Jin, F. F., Min, S. K., & Kam, J. (2026). Local and remote drivers of increased variability in extreme wildfire conditions in Southeastern Australia. *Agricultural and Forest Meteorology*, 384, 111156.
- [12] Lee, S., Lee, M. I., Lee, S., Jin, F. F., **Choi, N.**, & Tak, S. (2025). Distinct Features of the Summer Arctic Oscillation. *Journal of Climate*, 38(21), 5983-5996.
- [11] **Choi, N.**, Lee, M. I., Ham, Y. G., Hyun, Y. K., Lee, J., & Boo, K. O. (2025). Reducing initialization shock by atmosphere-ocean coupled data assimilation and its impacts on the subseasonal prediction skill. *Journal of Climate*, 38(6), 1389-1401.
- [10] **Choi, N.**, Lee, M. I., Tak, S., Cha, D. H., & Watanabe, M. (2025). Siberian heat extremes caused by Eurasian atmospheric teleconnections and amplified by local land surface conditions. *Environmental Research Letters*, 20(3), 034029.
- [9] **Choi, N.**, & Stan, C. (2025). Large-Scale Surface Air Temperature Bias in Summer over the CONUS and Its Relationship to Tropical Central Pacific Convection in the UFS Prototype 8. *Journal of Climate*, 38(1), 117-129.
- [8] Kim, H., **Choi, N.**, Lee, M. I., Tak, S., Cha, D. H., & Min, S. K. (2024). Causes on the 2018 record-breaking heatwave in South Korea: compound impacts from recurrent large-scale atmospheric teleconnections. *Environmental Research Letters*, 19(12), 124064.
- [7] Tak, S., **Choi, N.**, Lee, J., & Lee, M. I. (2024). Probabilistic medium-range forecasts of extreme heat events over East Asia based on a global ensemble forecasting system. *Weather and Climate Extremes*, 45, 100694.
- [6] **Choi, N.**, & Lee, S. (2023). Comparison of the opposite behaviours of Korean heatwaves with extreme hot sea surface temperatures in August 2016 and 2022. *International Journal of Climatology*, 43(15), 6993-7002.
- [5] Lee, S., Park, M. S., Kwon, M., Park, Y. G., Kim, Y. H., & **Choi, N.** (2023). Rapidly changing East Asian marine heatwaves under a warming climate. *Journal of Geophysical Research: Oceans*, 128(6), e2023JC019761.
- [4] Yasunari, T. J., Nakamura, H., Kim, K. M., **Choi, N.**, Lee, M. I., Tachibana, Y., & da Silva, A. M. (2021). Relationship between circum-Arctic atmospheric wave patterns and large-scale wildfires in boreal summer. *Environmental Research Letters*, 16(6), 064009.
- [3] **Choi, N.**, Lee, M. I., Cha, D. H., Lim, Y. K., & Kim, K. M. (2020). Decadal changes in the interannual variability of heat waves in East Asia caused by atmospheric teleconnection changes. *Journal of Climate*, 33(4), 1505-1522.
- [2] **Choi, N.**, Kim, K. M., Lim, Y. K., & Lee, M. I. (2019). Decadal changes in the leading patterns of sea level pressure in the Arctic and their impacts on the sea ice variability in boreal summer. *The Cryosphere*, 13(11), 3007-3021.
- [1] **Choi, N.**, & Lee, M. I. (2019). Spatial variability and long-term trend in the occurrence frequency of heatwave and tropical night in Korea. *Asia-Pacific Journal of Atmospheric Sciences*, 55, 101-114.

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- [4] Choi et al., Evolution of Boreal-Summer Quasi-Stationary Rossby Waves on Daily Time Scales. *npj Climate and Atmospheric Science*. Submitted.
 - [3] Tak et al., Incremental Contribution of Soil Moisture Data Assimilation to Sub-seasonal Prediction of the 2022 Central Eastern China Heatwave. *Weather and Climate Extremes*. In Review.
 - [2] Sohn et al., K-R2O in Action: Enhancing Operational Reliability through the Independent Testbed for National Climate Prediction System. *Bulletin of the American Meteorological Society*. In Review.

- [1] Xavier, P., Willett, M., ..., **Choi, N.**, ..., and Zhu, H.: Assessment of the Met Office Global Coupled model version 5 (GC5) configurations, Geoscientific Model Development Discussion. In Review.

PATENT

- Medium-Range Heatwave Forecasting System and Method (Domestic, South Korea, 2019)

AWARDS

- Young Atmospheric Scientist Award, Korean Meteorological Society (2025)
- Outstanding Poster Presentation Award, Korean Meteorological Society (2019)